Food Quality and Preference 144 (2026) 105987



Contents lists available at ScienceDirect

Food Quality and Preference

journal homepage: www.elsevier.com/locate/foodqual





Neuro-behavioral influence of ethnocentrism on beer consumers from different countries residing in Costa Rica

Silvia Campos-Esquivel [a], Pedro Baena-Luna [b,*], Pedro R. Palos-Sánchez [b]

[a] Instituto Tecnológico de Costa Rica, San José, Costa Rica
[b] Universidad de Sevilla, Seville, Spain

ARTICLE INFO

Keywords:
Neuromarketing
Ethnocentrism
Neuro-ethnocentrism
Consumer ethnocentrism
CETSCALE

ABSTRACT

Consumer ethnocentrism might influence purchasing decisions due to people's affinity for a brand's country of origin. With the rapid growth of digital technologies and e-commerce, it is crucial to determine whether ethnocentrism presents an opportunity or a barrier for firms entering foreign markets. The study is based on a non-probabilistic sample of 146 participants from four nationalities residing in Costa Rica, enabling exploratory analysis of neurobehavioral consumption patterns. Biometric technologies were used to measure attentional and emotional responses, attribute associations, and brand evaluation. Consumer ethnocentrism was assessed using a combination of eye-tracking, facial coding, implicit attribute-association testing, and the CETSCALE. Biometric data were processed using specialized analytical software, and Partial Least Squares Structural Equation Modeling was employed to validate the proposed model. The results suggest differentiated patterns of ethnocentric tendencies across national groups within the sample, with lower levels observed among Costa Rican and U.S. participants and comparatively higher levels among Mexican and Nicaraguan participants. The findings show consistency between experimental and traditional measures within the scope of this study, providing clear guidance for brands seeking to enter international markets.

1. Introduction

Firms continually try to enter foreign markets (Freixanet et al., 2020; Thornton & Zhao, 2025). Nevertheless, many of these firms encounter barriers, and only those that understand current global economic trends can survive, develop, and thrive (Lecerf & Omrani, 2020; Samiee et al., 2005). Even well-established brands have had difficulties establishing themselves in certain countries with great customer potential, such as China, India, Russia, and Brazil (Johansson & Leigh, 2011).

As Sorensen (2016) notes, the internationalization of a foreign company in a new market presents challenges related to language and cultural adaptation, as well as the skills required to gain acceptance, making it an intense and stressful process. Companies that successfully achieve this do so because they understand local consumers and their thinking, and can establish shared values (Zdravković & Gašević, 2022).

One key concept explaining consumer resistance to foreign brands is consumer ethnocentrism (Sun et al., 2021), which refers to the preference for domestic products over foreign alternatives (Rambocas & Mahabir, 2021) and for products from their own country over those from any other (Karoui & Khemakhem, 2019). Ethnocentrism, various social, psychological, economic, political, and demographic variables converge (Bizumic, 2019), with mediating factors such as equity, empathy, cost, responsibility, and country of origin (Shankarmahesh, 2006a, 2006b). Josiassen et al. (2011) highlight the roles of factors such as age and sex, as well as others more closely linked to consumption and socioeconomic conditions. At its core, ethnocentrism rests on individuals' belief that they should support and protect the national economy by purchasing domestically manufactured products (Balabanis & Diamantopoulos, 2004). Although previous studies have examined ethnocentrism in consumption, this area of research remains underexplored. Prior research has primarily relied on traditional qualitative and quantitative approaches, which may provide a limited or surface-level understanding of the underlying decision-making processes (Balabanis & Siamagka, 2022). Consequently, specific ethnocentric dilemmas—such as the choice between domestic and foreign products or the role of country of origin in shaping perceived value and quality during purchase decisions- remain insufficiently explored (Acharya & Elliott, 2003; Rašković et al., 2019). Moreover, the neurobehavioral mechanisms by which attentional and emotional responses, as well as value attribution, are shaped by reflective brand-preference models

* Corresponding author.
E-mail addresses: scampos@fundatec.ac.cr (S. Campos-Esquivel), pbaenaluna@us.es (P. Baena-Luna), ppalos@us.es (P.R. Palos-Sánchez).

https://doi.org/10.1016/j.foodqual.2026.105987
Received 9 October 2025; Received in revised form 28 April 2026; Accepted 17 May 2026
Available online 23 May 2026
0950-3293/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/).

incorporating ethnocentrism have not yet been empirically examined.

To address these gaps, the influence of ethnocentrism among beer consumers from four nationalities: Costa Ricans (as nationals) and Nicaraguans, U.S. residents, and Mexican residents (as foreigners), in the metropolitan area of Costa Rica is examined. Three Research Questions (RQ) guide the study:

- RQ1: What are the implicit attribute-associations and emotional reactions to brands because of the ethnocentrism of the consumer?
- RQ2: What is the weighting of price choice in the neuro-decisive process when buying different brands?
- RQ3: What are the structural models for assessing the degree of consumer ethnocentrism?

To address these research questions, a study was conducted using quantitative data collected from 146 participants from four nationalities who had been residing in Costa Rica for at least three months. A hybrid methodological approach integrating multiple quantitative techniques was employed to yield robust, multidimensional insights into consumer decision-making processes across cultural contexts. Biometric technologies, including eye-tracking and facial emotion-detection algorithms, were used to capture attentional and emotional reactions (Van Loo et al., 2019). The experimental design accounted for individual differences in consumer ethnocentrism, and implicit attribute-associations between product attributes and brand perceptions were examined. Consumer ethnocentrism was measured using the well-established CETSCALE (Shimp & Sharma, 1987).

By integrating biometric indicators, such as eye-tracking data, facial expression coding, and implicit attribute-association tests, with traditional psychometric measures, the study provided a nuanced understanding of how consumers across countries process, evaluate, and respond to brand-related stimuli. This methodological combination not only validated the proposed hypotheses but also demonstrated the relevance of hybrid quantitative models for consumer research. The findings provide exploratory insights into how levels of consumer ethnocentrism interact with emotional and attentional responses to influence brand perception and choice, thereby contributing to a more comprehensive understanding of cross-cultural consumer behavior in increasingly globalized markets. Consumer ethnocentrism has been extensively studied, as documented in previous integrative and systematic reviews (Shankarmahesh, 2006a, 2006b; Trivedi et al., 2024). Consequently, this study's contribution does not lie in the novelty of the construct, but rather in its application to a specific category (beer) and in the integration of implicit and explicit measures in a multicultural context, an aspect that has been less explored in existing literature.

2. Theoretical framework

2.1. Ethnocentrism in consumption

Bizumic (2019) demonstrates how territorial culture generates and reinforces the characteristics of different human groups, such as those from other tribes, towns, cities, and countries, thereby accentuating differences between them and giving rise to distinct groups of individuals.

Ethnocentrism is a phenomenon characterized by attitudes that make individuals more self-centered and self-important due to group affiliation (Poelmans et al., 2022). Recent work has employed a technological approach to group cohesion (Palos-Sanchez et al., 2023). The phenomenon of ethnocentrism has been addressed in scientific literature, including both a pessimistic analysis of the possible damage to one's own culture and generating hostilities between groups (Cunningham et al., 2004; Etinson, 2018; Livneh, 1982) and a positive approach due to the ability to create a feeling of belonging to a group and feeling proud (Bizumic, 2019).

The concept of consumer ethnocentrism (CE) originated in Shimp & Sharma (1987) research, which examined protectionism in the consumption of goods by the United States population. The research introduced and defined CETSCALE, a measurement instrument to assess consumer ethnocentrism. CE is an application of ethnocentrism to the economic sphere, from which the main premises and properties of this reality are derived (Balabanis & Diamantopoulos, 2004). Many scientific studies have used the scale to investigate how consumers prefer products from their own country of origin, even when they do not perceive them as superior (Weber et al., 2018).

The CE is a challenge for firms seeking to introduce products into foreign markets and can be a barrier to entering new international markets (Shimp & Sharma, 1987). It is a barrier that, although not price-related, further segments the potential market (Rašković et al., 2019). Maksan et al. (2019) apply the Theory of Planned Behavior, as identified by Ajzen (1991), to demonstrate how CE significantly influences attitudes toward purchasing and how it can alter a previously acquired planned attitude or buying habit.

As Kilders et al. (2021) point out, the influence of consumer ethnocentrism is not uniform and can vary significantly across product categories, levels of symbolic involvement, and links to cultural identity. In the context of this research, beer is a particularly relevant category, as it combines functional attributes with high symbolic, social, and identity content, especially in markets such as Costa Rica, where local beer consumption has historically been associated with social practices, traditions, and a sense of belonging.

Likewise, analysis of a multicultural population, Costa Rican consumers and Nicaraguan, US, and Mexican residents, allows us to observe how ethnocentrism operates differently depending on the degree of cultural roots, residence, and exposure to local brands, thereby reinforcing the relevance of studying this specific category. In line with Kilders et al. (2021), it is argued that ethnocentrism exerts a greater influence on frequently consumed and culturally familiar products, such as beer, where the product's origin can trigger both implicit responses and stated attitudes that influence purchase decisions.

In many cases the literature about consumer ethnocentrism explains the ethnocentric ideological estimation as a response to emotions that directly influence the consumer's reactions, motivated by a possible implicit emotional relationship with the product of the country of origin and aberration by the foreigner (Sharma, 2015), the effect of possible nostalgia for one's place of origin (Gineikiene & Diamantopoulos, 2017; Heinberg et al., 2020), the consumers' emotional involvement with the product (Rašković et al., 2019) and analyze the emotional congruence of advertising, pointing out that, if it is positive, it increases the fluency of cognitive processing and facilitates the management of information in an advertisement (Septianto et al., 2020).

High levels of consumer CE for a product influence variables that shape consumers' decision-making (Balabanis & Siamagka, 2022). These variables are Brand Awareness (BA), Quality Awareness (QA), Price Awareness (PA), and Information Usage (IU).

BA is the part of the consumer's personality that guides mental orientation when selecting a product based on brand awareness (Kautish et al., 2020). The degree of ethnocentrism a consumer exhibits toward a brand can affect their knowledge and understanding of highly advertised brands, which, in turn, can influence their relationship with the brand (Haikal, 2018). The following hypothesis is established using this idea:

H1. High levels of ethnocentrism among beer consumers have a direct and positive effect on the product's brand awareness.

Consumers' standards and expectations for a specific product will influence their QA (Ngoc et al., 2022). This means that a high-quality product is usually associated with a high price and a well-known brand (Balabanis & Siamagka, 2022). High levels of CE may influence this perception of QA (Acharya & Elliott, 2003). The following hypothesis is established using this information:

H2. High levels of ethnocentrism among beer consumers have a direct and positive effect on consumers' quality consciousness.

Consumers' high perceived value increases their confidence when buying a specific product. This relationship confirms the relationship between perceived value and customer trust (Ngoc et al., 2022). The customer's devotion to and desire for products of their nationality can affect their willingness to pay for them, thereby influencing the PA (Acharya & Elliott, 2003). The money spent on the purchase, including the product price, market offer, and potential price reductions, influences price-based purchasing decisions and is included in this variable. The following hypothesis is therefore proposed:

Hypothesis 3. High levels of ethnocentrism among beer consumers have a direct, positive effect on price awareness.

Information, today more than ever, is accessible to the millions of consumers who make purchases across different media. The amount of information a consumer has about a product or service can significantly influence their choice and perception of it (Stere & Trajani, 2015). High levels of CE can favor the biased use of information when making a purchase decision (Siamagka & Balabanis, 2015), which is the basis for the following hypothesis:

H4. High levels of consumer ethnocentrism have a direct and positive effect on the use of information about beer consumption.

2.2. Neuromarketing

Neuromarketing is a concept that combines neuroscience and consumer psychology, utilizing neurophysiological tools to promote firms' commercial interests (Javor et al., 2013).

It is especially relevant today due to a lack of confidence in traditional research, which leads to biases and harmful practices, such as researchers' subjective interpretation of consumers' implicitly subjective responses (Alsharif et al., 2021; Dovas et al., 2017). The reference to greater “subjectivity” is not intended to disqualify these approaches, but rather to point out that self-reported methods depend on the participant's ability to verbalize attitudes, emotions, or motivations, which can be affected by biases such as social desirability or ex post rationalization (Kahneman, 2014; Nisbett & Wilson, 1977).

The scientific community agrees that incorporating biometric measurement technologies is essential to successfully exploit the potential of neuromarketing (Braeutigam et al., 2004; Pieters, Warlop, & Wedel, 2007; Pieters, Wedel, & Zhang, 2007). These tools provide objective information about consumers' attention, emotions, and cognitive processes, complementing traditional market research methods and offering a deeper understanding of purchasing behavior. In this regard, various authors point out that biometric tools—such as eye tracking, galvanic skin response, EEG, or facial coding—provide valuable information about implicit, automatic, and non-conscious processes that are not always accessible through traditional declarative methods (Ariely & Berns, 2010; Plassmann et al., 2015). However, the literature also recognizes significant challenges, including the interpretive complexity of physiological signals, the need for rigorous experimental design, and the impossibility of completely replacing traditional methods (Ramsøy, 2019; Venkatraman et al., 2015).

According to Lee et al. (2018), biometric technologies can be classified into three broad categories. The first includes those that directly or indirectly record brain activity, providing data on the neural processes underlying decision-making. The second comprises technologies that directly or indirectly measure extracerebral or bodily neural activity, capturing physiological signals and emotional expressions that reflect cognitive and affective reactions. The third category encompasses methods that manipulate brain activity, typically for experimental or therapeutic purposes.

In this study, technologies from the second group were used, specifically eye tracking (ET) and facial coding (FC). These tools enable accurate, non-invasive analysis of participants' visual attention and emotional expressions, providing reliable indicators of how consumers perceive, process, and respond to brand and product stimuli (Lewinski et al., 2014; Wedel & Pieters, 2008a, 2008b). Their application is essential for understanding consumer behavior patterns that are not always evident through traditional research techniques.

ET technology measures consumers' conscious and unconscious attentional responses during purchasing decision-making (Wedel & Pieters, 2008a, 2008b). It is used to collect statistical information about consumers' purchase decisions (rational or unconscious).

FC is a technology that detects facial expressions and classifies them as fundamental and/or universal. The accuracy and reliability of these measurements depend on multiple factors, such as the hardware used, experimental conditions, processing algorithms, and the context of use, as evidenced by studies, such as that by Feit et al. (2017), in the case of eye tracking.

These tools do not replace traditional methods, but rather complement them, providing additional indicators of attentional and emotional processes that may not be fully accessible through declarative techniques. In this sense, their value lies in facilitating methodological triangulation, rather than offering absolute or universally accurate measures of consumer behavior (Plassmann et al., 2015; Ramsøy, 2019).

The use of these two techniques allows the following elements of the person's consumption decision to be measured with the Attentional Reaction (AR), which are the responses of the visual cortex made by the consumer to specific stimuli (Kahneman, 2014), the Emotional Reaction (ER) which is the emotional link between a brand or product and the consumer, from which the amount of preference for the origin or nationality can be measured. This is used to measure stereotypes about different groups based on race, age, gender, and other factors (Kurdi et al., 2021). Value Choice (VC) is the association of positive and negative attributes to a product, which establishes the relationship between the value that the consumer gives to a product and what they are willing to pay for it compared to the actual sale value of the product in the market (Totzek & Jurgensen, 2020).

The influence of CE on the variables of the consumption decision by the consumer is used to establish the following hypotheses:

H5. High levels of ethnocentrism among beer consumers have a direct positive effect on their attentional responses, as reflected in greater visual attention and increased cognitive processing of stimuli.

H6. High levels of ethnocentrism among beer consumers have a direct positive effect on consumers' emotional responses, as reflected in a stronger, more favorable emotional connection with the product.

H7. High levels of ethnocentrism among beer consumers have a direct positive effect on consumers' value perceptions, reflected in a higher perceived product value.

3. Methodology

3.1. Data collection and sample

The sample for this research consisted of 146 participants, both male and female, who had resided in Costa Rica for more than three months. The sampling for this research is non-probabilistic and based on convenience and the availability of informants. Individuals were selected based on criteria of having defined consumption patterns and significant exposure to local brands, in accordance with recommendations from previous studies on consumer behavior (Weber et al., 2018). The first consumption pattern is that they will claim to consume beer, that their residence in the country for at least 3 months ensures exposure to the local brand ‘Imperial’, and that they have been exposed to this brand. Four brands are included: one is national (Costa Rica's ‘Imperial’), and the other three are imported and share the nationality of the subjects evaluated (Nicaragua's ‘Toña’, the United States' ‘Miller’, and Mexico's

'Corona'.

People from Nicaragua living in Costa Rica do not have a different profile. Still, they are exposed to local brands such as Imperial, whereas Nicaraguans living in Nicaragua are not.

The sample is representative, and the minimum size of each subsample was obtained, so that the subsample of the Costa Rican population was calculated from a population under study determined by inhabitants of the greater metropolitan area, of a medium and higher socioeconomic level, with fixed incomes and beer consumers (48.3%), yielding a total of 411,101. The minimum subsample of the foreign population was determined based on the three nationalities studied, yielding a total of 382,617. Thus, both populations required a minimum subsample of 72.98; $n = 146$ individuals were studied. The rest of its features were anonymous to respect your privacy. The final distribution by nationality included 25 Nicaraguans, 24 Americans, 24 Mexicans, and 73 Costa Ricans. However, it is important to note that the subsample sizes by nationality are relatively small, which limits the robustness of multigroup comparisons.

Given that the study focuses on decision-making and purchasing choices, only adults aged 18 to 70 with the authority to purchase various beer brands were included. Additionally, participants had to meet specific health criteria to ensure the validity of the neurobehavioral measurements, including normal or corrected vision, no use of psychoactive medications, and no history of neurological or psychiatric disorders. These restrictions ensured that the collected data reflected genuine behavior patterns and were not influenced by external factors.

Participant selection was conducted by two recruitment firms, ensuring that all participants met the required profile. Various experimental tests were performed in accordance with standardized protocols for collecting biometric and neuromarketing data. To comply with Articles 4 and 5 of the World Neuromarketing Science and Business Association's (NMSBA) Code of Ethics, all participants signed an informed consent form that included their name, date, and signature, confirming their voluntary participation in the study.

The statistical methodology, based on Partial Least Squares Structural Equation Modeling (PLS-SEM), was selected for its suitability to complex models and its justified use in similar studies (Reyes-Menendez et al., 2022), thereby enabling the analysis of relationships among latent variables and the validation of the proposed hypotheses. Given the exploratory nature of the study and the use of Partial Least Squares Structural Equation Modeling (PLS-SEM), the sample size ($n = 146$) is considered adequate for estimating the global structural model. PLS-SEM is particularly suitable for exploratory research and predictive modeling in complex frameworks (Hair et al., 2013). Therefore, the results should be interpreted within the context of the sample and not as statistically generalizable to the broader population.

However, following the recommendations of Hair et al. (2013) for determining the minimum sample size for the PLS model, Cohen's (1992) tables were used. To apply these tables accurately, following previous studies (Saura, Palos-Sanchez, & Blanco-González, 2020; Saura, Palos-Sánchez, & Velicia-Martín, 2020), we were supported by the G*Power software (Faul et al., 2009). First, we verified the existence of dependent constructs. Among them, we chose the one with the largest number of predictors; in our case, both Intent to Use and Brand, Price, and Quality Awareness have a single predictor. Therefore, as can be seen in Fig. 1, we evaluated the parameters: power of the test (power = 1 - β probability of error II) and effect size (f^2). Following the recommendations of the mentioned authors: power = 0.80, predictors = 1, and a mean effect size $f^2 = 0.15$. As a result, for multiple linear regressions, the accepted number of participants in the sample was a minimum of 55 (see Fig. 1), which we exceeded by using $n = 117$.

Overall, the sample design and procedures used ensure the robustness and reliability of the results, providing relevant information on consumption patterns and the influence of cultural factors on consumer decision-making. The matrix containing the imported data derived from the research, the eye tracker, and facial coding analysis software, as well

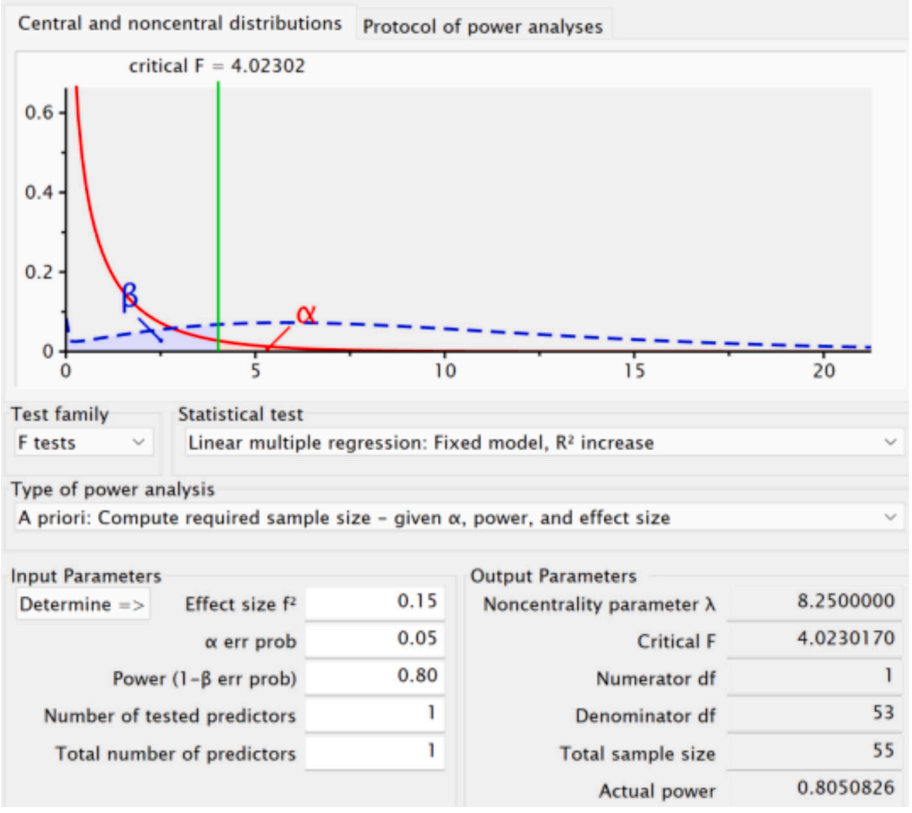


Fig. 1. Sample size calculation.
Source: Authors' own elaboration.

as the survey results, was exported to Excel. Using this information, the reliability and validity of the measurement instruments were evaluated.

3.2. Instruments

This study employs a quantitative, experimental, and cross-sectional design; it is analytical, deductive, synthetic, and nomothetic, combining traditional procedures with innovative neuromarketing techniques and focusing on an area with limited prior research: the impact of ethnocentrism on consumer behavior.

This quantitative research combines an experimental design in the first stage and a correlational design in the second, with participants completing the CETSCALE (Shimp & Sharma, 1987). It is considered cross-sectional, as it collects data at a single point in time (Demirtaş & Altuntaş, 2024). The two main instruments of this research, the neuromarketing experiment and the questionnaire, were applied.

In this research, the operational map comprises two processes: prior recruitment, which corresponds to field preparation and the first stage of data collection using biometric technologies; and the second stage, in which the CETSCALE survey is administered. The study adopts a two-stage, sequential research design to capture complementary levels of the decision-making process associated with consumer ethnocentrism. In the first stage, neuromarketing technologies are applied to evaluate implicit, automatic, and non-conscious responses to stimuli related to product origin, which aligns with System 1 processing as described by Kahneman. This approach reduces social desirability bias and subsequent rationalizations. In the second stage, CETSCALE is applied to measure explicit, deliberate, and self-reported attitudes of consumer ethnocentrism, corresponding to System 2 processing, and to facilitate standardized comparison between cultural groups.

This methodological sequence strengthens the study's robustness by integrating implicit and explicit measures, in line with the literature, which indicates that ethnocentrism influences decision-making at both conscious and unconscious levels.

The scale developed by Rašković et al. (2019) uses an expanded version of the CETSCALE to more accurately measure consumer ethnocentrism among young adults from diverse cultural backgrounds in East Asia and Central and Eastern Europe. Given that this construct can vary significantly across sociocultural environments, expanding the scale improves the measurement's sensitivity and validity, as well as its ability to reflect differences across countries. This adaptation is particularly relevant for examining the relationship between consumer ethnocentrism and decision-making styles, thereby ensuring more robust, contextually grounded measurement. No other instruments or measures, such as the CSI, were used; the study was limited to the CETSCALE.

3.2.1. Biometric measurement with eye tracker

Eye tracking is a technology that records neural activity outside the brain. This research uses Tobii eye-tracking technology, model x2–60, with a sampling rate of 60 Hz (approximate standard deviation of 0.1 Hz) and a head movement range of 50 × 36 cm (20 × 14 in.) at a width x height of 70 cm.

Tobii Pro Lab software version 1.171.34906 (x64) is used for information processing. This comprehensive platform enables the recording and analysis of visual data and facilitates the study and interpretation of human behavior, consumer responses, and psychological processes (Tobii Pro AB, 2019). It collects consumer information in response to visual stimuli, using timestamps, eye position, gaze points, pupil diameter, and validity code.

3.2.2. Biometric measurement with facial coding

For the analysis of information using facial coding technology, Noldus Facial Coding TM software, version 8.0, was used. Facial coding is a program that can detect facial expressions. It classifies expressions into one of the following categories: happy, sad, angry, surprised, scared, disgusted, and neutral. These emotional categories have been described by Ekman (1970) as basic or universal. In addition to the emotion type, this algorithm supports valence and arousal calculations, action unit classification, and analysis of the subject's characteristics.

The emotional impact of an individual or group is generally a function of valence and arousal level, as measured when subjects are exposed to the stimulus. These variables, according to (Loijens & Krips, 2021), are defined as:

Valence: Indicates whether the subject's emotional state is positive or negative. For example, suppose the happiness curve (the only positive emotion) is 0.8 and the sadness, anger, fear, and disgust curves are 0.2, 0.0, 0.3, and 0.2, respectively. In that case, the equation for generating valence is 0.8–0.3 = 0.5.

Arousal level: Indicates whether the participant is active (+1) or inactive (0). Arousal is based on the activation of 20 action units (AU) of the Facial Action Coding System (FACS) (Ekman et al., 2002).

3.2.3. CESTCALE Instrument

The original CETSCALE survey, designed to measure ethnocentrism among consumers, comprises 17 items and uses a 5- or 7-point Likert scale. Sharma (2015) assessed the scale's validity and dimensionality using cross-cultural measurement invariance, concluding that the scale applies to any subject, regardless of nationality, and to groups comprising subjects of multiple nationalities. In the present study, Rašković et al. (2019) proposed a modified version of the CETSCALE that incorporates the variables of interest. This instrument has 33 items, including six filter questions and 27 measured on a 5-point Likert scale, where 1 = strongly disagree and 5 = strongly agree.

Since filter questions are a tool for adapting to heterogeneous subpopulations (Alvira, 2011), four filter questions are included to ensure the homogeneity of the sample: nationality, as it only applies to subjects of four nationalities that share the nationality of the brands; age, since the majority of this is required for consumption of the beer category; beer consumption, to ensure knowledge of the brand/product; and finally, being the person who has the purchasing choice over brands in these categories, that is, being the purchasing decision-maker in their household.

3.2.4. Implicit attribute-association task

This study employed an implicit attribute-association task, supported by eye-tracking technology, to infer associative processes from patterns of visual attention. The experimental procedure consisted of a sequential presentation of stimuli in two stages. First, participants were exposed to attribute-based stimuli (e.g., positive or negative product-related attributes) displayed briefly on the screen. These attributes were presented for approximately 300 milliseconds, a duration consistent with pre-conscious or early-stage cognitive processing.

Immediately following this, brand stimuli corresponding to different national origins were presented for approximately 3 s. During this phase, participants were allowed to freely explore the visual stimulus without any explicit categorization task.

The underlying assumption of this task is that prior exposure to attributes may activate associative networks that influence subsequent visual attention patterns toward brands. In this context, shorter Time to First Fixation (TTFF) and longer Total Visit Duration (TVD) toward specific stimuli are interpreted as indirect indicators of implicit associative congruence.

It is important to note that, unlike the classical IAT, this procedure does not involve forced categorization, reaction-time comparisons between congruent and incongruent blocks, or the computation of D-scores. Therefore, the results derived from this task should be interpreted as exploratory indicators of implicit attribute-brand associations inferred from visual attention, rather than as direct measures of associative strength in the traditional IAT sense (Fazio & Olson, 2003; Kurdi et al., 2021).

4. Results

After the data processing stage, the proposed model and its results were analyzed in different ways. The composite reliability values (which measure internal consistency) ranged from 0.821 to 0.948 (Table 1). These values exceed 0.70, the minimum acceptable value for adequate reliability; therefore, each measurement accounts for at least 50% of the variance of the underlying construct (Henseler et al., 2009).

The Cronbach's Alpha values ranged from 0.643 to 0.936, exceeding the minimums of 0.6 (Chin, 1998) and 0.7 (Nunnally, 1978), which are typically recommended for exploratory studies like this one. The composite reliabilities exceeded 0.7 in all cases, and the Average Variance Extracted (AVE) for the constructs exceeded 0.5; therefore, convergent validity is acceptable (Sarstedt et al., 2014).

The discriminant validity (Henseler et al., 2015) of the structural equation models was assessed using variance values, the Fornell & Larcker (1981) criterion, and cross-loadings. This criterion requires that the metric's indicator be at least 0.50, indicating that the variables explain more than half of the variance in their respective indicators. As shown in Table 2, the square root of the AVE is higher than its correlation with the other constructs. Note that RA equals 1.0 because it has only one item.

Both the Fornell & Larcker (1981) criterion and the heterotrait-monotrait matrix (HTMT) were used to assess discriminant validity, as the latter addresses issues related to the Fornell-Larcker criterion and cross-loadings (Rey-Moreno et al., 2018; Saura, Palos-Sanchez, & Blanco-González, 2020; Saura, Palos-Sánchez, & Velicia-Martín, 2020). The HTMT values in Table 3 indicate that the maximum matrix value is 0.317, which is well below the maximum acceptance limit of 0.85. These evaluations confirm the model has discriminant validity (Kaldeen & Sabraz Nawaz, 2019).

After validating the measurement model, the structural model was evaluated to assess the proposed relationships between EC and the study's constructs. After establishing the reliability and validity of the complete model, the Path Coefficients and the Determination Coefficient (R^2) were calculated (Becker et al., 2023).

To obtain the indicator factor loadings and significance, the structural model was analyzed using bootstrapping with 5000 subsamples (Becker et al., 2023; Martínez-Ávila & Fierro-Moreno, 2018). The detailed results for the trajectory coefficient (β), t-statistics, and *p*-values are presented in Table 4 and Fig. 2. To determine the statistical significance of the path coefficients, the same bootstrapping technique was applied based on an original sample of 146 participants.

The results confirm that CE has a significant and positive influence on four of the seven constructs analyzed.

The Coefficient of Determination (R^2) indicates the proportion of the variance in the endogenous constructs explained by the model. The model shows its greater predictive capacity for Intention to Use (IU). The predictive performance of the PLS-SEM sampling criteria is demonstrated by the R^2 value (Chin, 1998). The coefficient of determination indicates the predictive value of the variables for CE (Hair et al., 2013), with the construct exhibiting the highest predictive power in the test model. The results obtained are shown in Table 5.

Table 1
Cronbach's Alpha, Composite Reliability (CR), Average Variance Extracted (AVE).

Constructs	Cronbach's alpha	CR	AVE
CE	0.936	0.948	0.695
BA	0.705	0.872	0.772
PA	0.675	0.821	0.605
VC	0.643	0.846	0.733
ER	0.931	0.923	0.800
IU	0.769	0.866	0.684

Source: Authors' own elaboration.

5. Discussion

The objective of this study was to analyze how consumers' ethnocentrism toward another country influences their perceptions and beer choices, using a hybrid approach that combines psychometric and neurobehavioral measures. The analysis of the results indicates that ethnocentrism mainly influences through cognitive and attentional processes, while its effect on emotional responses to beer as a product is limited. This pattern is consistent with the findings of authors such as Fernández-Ferrín et al. (2018), who note that the effects of origin and national identity do not always translate into strong affective responses, especially in everyday consumption categories.

A first relevant finding of this study is that the relationship between Consumer Ethnocentrism and Price Awareness, as well as the selective use of information during decision-making, is extremely strong. This is because consumers with higher levels of ethnocentrism tend to give greater weight to functional and economic signals, processing information in a biased manner in favor of products from their country of origin. This result aligns with the findings of Shin et al. (2025), who show that ethnocentrism influences the valuation and purchase of food products with local, regional, or traditional attributes, thereby affecting real economic decisions, such as willingness to pay and preferences for domestic products over imported ones.

Another equally important aspect is how the results show a dissociation between visual attention and emotional response. Ethnocentrism had a significant effect on visual attention to beer brands, as measured by eye-tracking, whereas no significant effect was observed on Emotional Reaction, as assessed by facial coding. This finding is particularly relevant to food and beverage research, as it suggests that consumers may visually prioritize products that align with their national identity without necessarily feeling greater emotional involvement. This evidence extends the findings of Van Loo et al. (2019), who showed that ethnocentrism modulates visual attention toward labels of origin and extrinsic cues on food products, influencing the final choice through attentional rather than affective mechanisms.

In our work, however, we found that ethnocentrism did not significantly influence perceived quality or product value. In the multicultural context analyzed, characterized by high exposure to both local and foreign brands, the perception of quality seems to depend more on product familiarity and prior market knowledge than on national origin (Maró et al., 2023).

Three of the proposed hypotheses are not supported by the data collected. In the case of a direct, positive relationship between high levels of Consumer Ethnocentrism and Quality Awareness, the results show that the relationship between the two elements is neither significant nor positive. This result contrasts with previous research, such as that of Acharya and Elliott (2003), which demonstrated that the area of origin positively influences perceptions of product quality. Their study suggests that consumers tend to associate geographical origin with perceived quality attributes, a finding not reflected in the present study's data.

Similarly, the results regarding the potential positive impact of high Consumer Ethnocentrism levels on the Price Awareness and Emotional Reaction of a given product, in this case, are not statistically significant. This finding contradicts the results of Totzek and Jurgensen (2020), who highlighted Emotional Reaction as a key determinant of perceived value and the price consumers are willing to pay. In the present study, however, high levels of Consumer Ethnocentrism do not appear to influence the formation of this perceived value, suggesting that the effect of origin or provenance is mediated by other contextual or cultural factors not included in the model.

The results for the relationship between Consumer Ethnocentrism and Emotional Reaction indicate that high levels of Consumer Ethnocentrism among consumers do not lead to the formation of product stereotypes based on their territorial origin. This result coincides with that reported by Kurdi et al. (2021), who argue that the growing

Table 2
Fornell-Larcker criterion.

	QA	CE	BA	PA	VC	AR	ER	IU
QA	1.000							
CE	0.133	0.834						
BA	0.530	0.199	0.879					
PA	0.130	0.319	0.222	0.778				
VC	−0.163	−0.132	−0.181	−0.162	0.856			
AR	0.088	−0.152	−0.127	0.071	−0.070	1.000		
ER	−0.031	−0.079	0.012	−0.085	−0.056	−0.080	0.894	
IU	0.213	0.499	0.273	0.242	−0.144	−0.053	0.011	0.827

Source: Authors' own elaboration.

Table 3
HTMT correlation matrix.

	QA	CE	BA	PA	VC	AR	ER	IU
QA								
CE	0.135							
BA	0.633	0.243						
PA	0.163	0.394	0.317					
VC	0.196	0.175	0.265	0.225				
AR	0.088	0.157	0.151	0.084	0.089			
ER	0.100	0.082	0.077	0.113	0.080	0.050		
IU	0.236	0.582	0.363	0.332	0.197	0.099	0.052	

Source: Authors' own elaboration.

Table 4
Path coefficients and statistical significance.

Hypothesis	β	t statistic	p-value	Supported
CE - > QA	1.427	1.432	0.153	No
CE - > BA	2.235	2.212	0.027	Yes *
CE - > PA	4.679	4.461	0.000	Yes ***
CE - > VC	1.844	1.701	0.090	No
CE- > AR	2.094	2.252	0.025	Yes*
CE - > ER	0.613	0.586	0.558	No
CE - > IU	7.522	7.340	0.000	Yes***

n = 5000 subsamples: * p < .05; ** p < .01; ***p < .001; ns: not significant (two-tailed t-Student); t (0.01. 499) = 2.333843952); (t (0.001; 499) = 3.106644601. Note: n = 500 refers to the number of bootstrap subsamples used in the PLS-SEM analysis (Sabol et al., 2023); the original sample consists of 146 participants.
Source: Authors' own elaboration.

globalization of consumption and the diversification of information channels tend to dilute traditional associations between provenance and quality stereotypes. Overall, the findings indicate that although some theoretical relationships are confirmed with high confidence, others weaken in contexts where consumers possess broader, less biased knowledge about the analyzed products.

This answers the first two research questions posed, RQ1 and RQ2. The data collected does not reveal direct positive relationships among beer consumers with high Consumer Ethnocentrism levels.

The following research question (RQ3) investigates the structural models used to assess the degree of ethnocentrism among consumers. The results clearly show that consumers with a high level of Consumer Ethnocentrism have greater brand awareness of certain products (Haikal, 2018; Kautish et al., 2020). The relationship between consumers' perceived value of beer and other products and their nationality may affect their willingness to pay higher prices (Acharya & Elliott, 2003). High levels of Consumer Ethnocentrism indicate that consumers tend to use information in a biased manner, favoring products from their own country of origin at the expense of those from other countries (Balabanis & Diamantopoulos, 2004; Balabanis & Siamagka, 2022). Similarly, in Attentional Reaction, the visual cortex responds positively and directly to high-level manifestations of Consumer Ethnocentrism (Kahneman, 2014).

Finally, although it was not an objective of the research, there is previous evidence derived from research related to this same project (Campos Esquivel & Palos Sánchez, 2023), which documents differences in neurophysiological responses to the origin of the product in beer consumers and shows variations in perceived value, associated with ethnocentrism, depending on the cultural context. However, it was not analyzed using multigroup Partial Least Squares Path Modeling because it did not meet one of the Measurement Invariance of Composite Models criteria: Compositional Invariance (Henseler et al., 2016; Sarstedt et al., 2011). As the minimum requirement for proceeding with Multigroup Analysis was not met, only univariate analyses are available.

The integration of biometric and psychometric measures enabled consistent triangulation of results, strengthening the internal validity of the observed patterns within the sample. However, these findings should be understood as exploratory and context-specific rather than definitive evidence of cross-national differences.

5.1. Theoretical implications

The results and analysis of this study have significant theoretical implications for understanding consumer behavior, particularly regarding the role of ethnocentrism in purchasing decisions. This research provides detailed evidence on how consumer preferences and choices are shaped by an affinity for domestic versus foreign products, thereby contributing to a deeper understanding of the cultural dynamics underlying consumer behavior. In this way, it expands existing theory, offering a more accurate account of the interaction among cultural identity, personal values, and purchasing decisions in multicultural contexts.

A central aspect of this work is the application of neurobehavioral techniques through biometric methods, which represents a significant advance over previous studies that relied exclusively on traditional qualitative or quantitative methods. The use of tools such as Eye Tracking and Facial Coding enables the objective capture of consumers' attention, emotional responses, and the value they associate with products. This not only facilitates the identification of ethnocentric behavior patterns but also provides information on the causal relationships that influence decision-making, an aspect that cannot always be addressed through conventional surveys or interviews.

Furthermore, this study's findings highlight the role of neurobehavioral factors in brand and product perception, showing that emotional and cognitive responses are closely linked to preferences for local versus foreign products. This enables researchers and firms to gain a deeper understanding of the factors that influence the acceptance or rejection of specific products in culturally diverse markets.

These methodological and conceptual innovations strengthen the theory of consumer ethnocentrism, offering a more robust framework for future research. By combining biometric approaches with ethnocentric behavior analysis, this study helps close a gap in the literature, providing a strong empirical basis for both academic contexts and the design of culturally sensitive and effective marketing strategies.

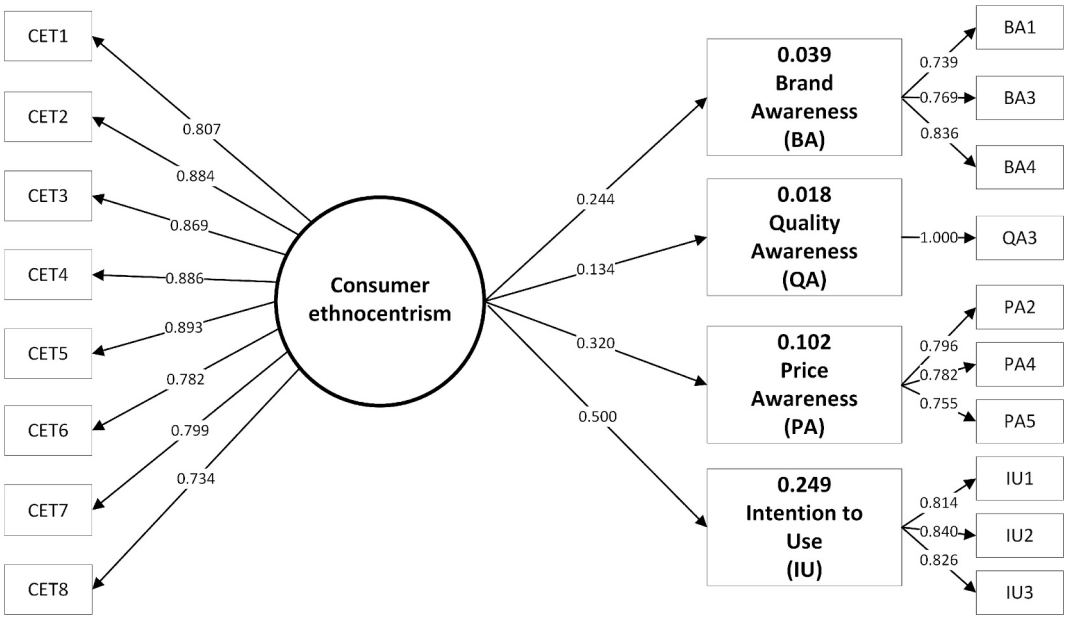


Fig. 2. Path modeling.
Source: Authors' own elaboration.

Table 5
The determination coefficient (R^2).

Constructs	R^2 squared
QA	0.018
BA	0.039
PA	0.102
VC	0.017
AR	0.023
ER	0.006
IU	0.249

Source: Authors' own elaboration.

5.2. Practical implications

Equally relevant are the practical implications of this research, particularly in the beer industry, where understanding consumer behavior is crucial for developing effective marketing and brand positioning strategies. The results of this study provide detailed insights that are highly useful to managers and decision-makers in beer-producing firms, enabling them to anticipate how linking a brand to a specific territory can influence consumers' purchasing decisions. In particular, the research indicates that consumer ethnocentrism can significantly influence preferences for local versus foreign products, suggesting that communication and positioning strategies must be tailored to the target market's cultural and social characteristics.

This knowledge becomes especially valuable in regions where the population and products originate from diverse cultural and geographical sources, which can lead to high levels of Consumer Ethnocentrism and affect brand perception. Identifying behavior patterns associated with Consumer Ethnocentrism enables firms to design more effective marketing campaigns, establish brand associations that highlight relevant cultural attributes, and, in some cases, reassess the brand's connection to specific territories or cultural symbols.

In addition, the application of neurobehavioral techniques with biometric tools provides a deeper, more objective view of consumers' emotional and attentional responses, allowing firms to understand not only which products are preferred but also why and how those preferences form at the cognitive and affective levels. This represents a strategic resource for anticipating consumer trends, adapting product offerings, and fostering a stronger emotional connection between the brand and consumers, thereby increasing loyalty and product acceptance across diverse markets.

These practical implications provide beer producers and other industry players with a valuable framework for designing culturally informed, evidence-based marketing strategies that optimize brand acceptance and consumer satisfaction, ultimately contributing to commercial success in a competitive and multicultural environment.

The findings from the implicit attribute-association task provide exploratory evidence that prior attribute exposure may influence subsequent visual attention to brands. By integrating eye-tracking metrics, the study offers an alternative approach to examining implicit associative processes, complementing traditional self-report measures.

However, it is important to clarify that the method used does not correspond to the classical Implicit Association Test (IAT) paradigm. Instead, the associations observed in this study are inferred indirectly from visual attention indicators such as TTFF and TVD. In line with existing literature, implicit measures can vary significantly in their underlying mechanisms and interpretive scope and should not be treated as interchangeable without explicit methodological justification (Fazio & Olson, 2003; Kurdi et al., 2021).

5.3. Limitations and future lines of research

This research is not exempt from certain limitations. The same applies to the study's territorial scope, which is limited to Costa Rica. This, along with the sample size, suggests caution when generalizing or extrapolating from the results.

First, the use of a non-probabilistic convenience sample limits the generalizability of the findings. Second, the relatively small subsample sizes by nationality restrict the ability to perform robust multigroup comparisons. Third, the PLS-SEM model was estimated on the combined sample and does not test measurement invariance or structural differences across groups. Future research should address these limitations by employing larger, more balanced samples, using probabilistic sampling strategies where feasible, and employing formal multigroup analysis procedures.

Future research based on this work could explore the relationship between ethnocentrism and beer consumption. Some possible future lines of research could address the main limitations of this study. A

longitudinal study would allow more accurate conclusions about the effect of ethnocentrism on beer consumption among people of different nationalities or countries of origin in Costa Rica. Including consumers from other countries in the research would enable more accurate causal inferences about the effect of ethnocentrism on product consumption.

6. Conclusions

Based on the results and discussion of this study, the most relevant conclusion is the importance of ethnocentrism as a real and significant phenomenon in consumer behavior. First, this study suggests that ethnocentrism is not an isolated theoretical concept but manifests itself in tangible ways, including product choices, such as beer. It affects people of different nationalities in different ways. This finding is particularly relevant, as it enables us to identify patterns of preference and rejection that are deeply rooted in cultural identity and a preference for domestic products over foreign ones, thereby providing a more transparent framework for understanding consumer behavior. Secondly, research has shown that integrating biometric techniques, such as eye tracking and facial coding, with traditional methods used in previous studies enables a more in-depth exploration of ethnocentrism. This methodological combination provides a more comprehensive analysis of the influences that ethnocentrism has on consumers, as it allows us to observe not only stated preferences but also the attentional and emotional responses that underlie decision-making. Thus, the findings provide concrete evidence on how consumers' cognitive and affective reactions relate to their levels of ethnocentrism. Finally, the results obtained indicate that high levels of ethnocentrism significantly influence consumers' perceptions of the brand and product price, as well as their use of available information during the purchase decision process. At the neurobehavioral level, apparent effects on attentional responses are also evident, suggesting that cultural predispositions affect not only conscious evaluations but also automatic cognitive and emotional processes.

The conclusions of our work highlight the importance of considering ethnocentrism as a key factor in the design of marketing and brand positioning strategies, particularly in international and multicultural markets. They also emphasize the importance of integrating traditional methods with biometric techniques to gain a more comprehensive understanding of consumer behavior and develop evidence-based approaches that enhance product acceptance and the effectiveness of commercial strategies.

Declaration of funding

The authors have no relevant financial or non-financial interests to disclose.

CRediT authorship contribution statement

Silvia Campos-Esquivel: Writing – original draft, Project administration, Methodology, Data curation, Conceptualization. **Pedro Baena-Luna:** Writing – original draft, Validation, Resources, Investigation, Conceptualization. **Pedro R. Palos-Sánchez:** Validation, Software, Methodology, Investigation, Formal analysis.

Ethics approval statement

Ethical approval is not applicable for this article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets used and analyzed during the present study are available, adequately anonymized, from the corresponding author upon reasonable request.

References

Acharya, C., & Elliott, G. (2003). Consumer ethnocentrism, perceived product quality and choice—An empirical investigation. *Journal of International Consumer Marketing, 15*(4), 87–115. https://doi.org/10.1300/J046v15n04_05

Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes, 50*(2), 179–211. https://doi.org/10.1016/0749-5978(91)90020-T

Alsharif, A. H., Md Salleh, N. Z., Baharun, R., Rami Hashem, E., & A.. (2021). Neuromarketing research in the last five years: A bibliometric analysis. *Cogent Business & Management, 8*(1), Article 1978620. https://doi.org/10.1080/23311975.2021.1978620

Alvira, F. (2011). *La encuesta: una perspectiva general metodológica* (Volumen 35). CIS.

Ariely, D., & Berns, G. S. (2010). Neuromarketing: The hope and hype of neuroimaging in business. *Nature Reviews Neuroscience, 11*(4), 284–292. https://doi.org/10.1038/nrn2795

Balabanis, G., & Diamantopoulos, A. (2004). Domestic country Bias, country-of-origin effects, and consumer ethnocentrism. *Academy of Marketing Science Journal, Winter, 200*(32), 80. http://search.ebscohost.com/login.aspx?direct=true&db=bth&AN=12876978&site=ehost-live.

Balabanis, G., & Siamagka, N. T. (2022). A meta-analysis of consumer ethnocentrism across 57 countries. *International Journal of Research in Marketing, 39*(3), 745–763. https://doi.org/10.1016/j.ijresmar.2021.12.002

Becker, J. M., Cheah, J. H., Gholamzade, R., Ringle, C. M., & Sarstedt, M. (2023). PLS-SEM'S most wanted guidance. *International Journal of Contemporary Hospitality Management, 35*(1), 321–346. https://doi.org/10.1108/IJCHM-04-2022-0474

Bizumic, B. (2019). Effects of the dimensions of ethnocentrism on consumer ethnocentrism: An examination of multiple mediators. *International Marketing Review, 36*(5), 748–770. https://doi.org/10.1108/IMR-04-2018-0147

Braeutigam, S., Rose, S. P. R., Swithenby, S. J., & Ambler, T. (2004). The distributed neuronal systems supporting choice-making in real-life situations: Differences between men and women in grocery choice detected using magnetoencephalography. *European Journal of Neuroscience, 20*(1), 293–302. https://doi.org/10.1111/j.1460-9568.2004.03467.x

Campos Esquivel, S., & Palos Sánchez, P. R. (2023). Influencia del etnocentrismo: un estudio de las reacciones neurofisiológicas del consumidor de cerveza. *Revista de Marketing y Publicidad, 7*, 73–103. https://doi.org/10.51302/marketing.2023.7233

Chin, W. W. (1998). The partial least squares approach for structural equation modeling. In *Modern methods for business research* (pp. 295–336). Lawrence Erlbaum Associates Publishers.

Cohen, J. (1992). A power primer. *Psychological Bulletin, 112*(1), 155–159. https://doi.org/10.1037//0033-2909.112.1.155

Cunningham, W. A., Nezlek, J. B., & Banaji, M. R. (2004). Implicit and explicit ethnocentrism: Revisiting the ideologies of prejudice. *Personality and Social Psychology Bulletin, 30*(10), 1332–1346. https://doi.org/10.1177/0146167204264654

Demirtaş, H., & Altuntaş, S. (2024). Nurses' competence levels in disaster nursing management in Turkey: A comparative cross-sectional study. *International Nursing Review, 71*(3), 556–562. https://doi.org/10.1111/inr.12829

Dovas, P., Florentino, L., Grieve, C., Hofkirchner, H., MacInnes, C., Matukin, M., … Steidl, P. (2017). *The market research revolution.* Neuromarketing Science and Business Association Publications Series.

Ekman, P. (1970). Universal facial expressions of emotion. In *, vol. 8. California mental health research digest* (pp. 151–158).

Ekman, P., Friesen, W. V., Hager, J. C., & F., A. H. (2002). Facial action coding system. In *Manual and Investigator's Guide.* Research Nexus.

Etinson, A. (2018). Some myths about ethnocentrism. *Australasian Journal of Philosophy, 96*(2), 209–224. https://doi.org/10.1080/00048402.2017.1343363

Faul, F., Erdfelder, E., Buchner, A., & Lang, A.-G. (2009). Statistical power analyses using G*power 3.1: Tests for correlation and regression analyses. *Behavior Research Methods, 41*(4), 1149–1160. https://doi.org/10.3758/BRM.41.4.1149

Fazio, R. H., & Olson, M. A. (2003). Implicit measures in social cognition research: Their meaning and use. *Annual Review of Psychology, 54*, 297–327. https://doi.org/10.1146/annurev.psych.54.101601.145225

Feit, A. M., Williams, S., Toledo, A., Paradiso, A., Kulkarni, H., Kane, S., & Morris, M. R. (2017, May). Toward everyday gaze input: Accuracy and precision of eye tracking and implications for design. In *Proceedings of the 2017 CHI conference on human factors in computing systems* (pp. 1118–1130).

Fernández-Ferrín, P., Calvo-Turrientes, A., Bande, B., Artaraz-Miñón, M., & Galán-Ladero, M. M. (2018). The valuation and purchase of food products that combine local, regional and traditional features: The influence of consumer ethnocentrism. *Food Quality and Preference, 64*, 138–147. https://doi.org/10.1016/j.foodqual.2017.09.015

Fornell, C., & Larcker, D. F. (1981). Structural equation models with unobservable variables and measurement error: algebra and statistics. *Journal of Marketing Research, 18*, 382–388. https://doi.org/10.2307/3150980

Freixanet, J., Rialp, A., & Churakova, I. (2020). How do innovation, internationalization, and organizational learning interact and co-evolve in small firms? A complex

systems approach. *Journal of Small Business Management, 58*(5), 1030–1063. https://doi.org/10.1111/jsbm.12510
Gineikiene, J., & Diamantopoulos, A. (2017). I hate where it comes from, but I still buy it: Countervailing influences of animosity and nostalgia. *Journal of International Business Studies, 48*(8), 992–1008. https://doi.org/10.1057/s41267-017-0097-5
Haikal, D. M. (2018). The effect of consumer ethnocentrism, brand image, and perceived quality on purchase decisions with purchase intention as intervening variable. *Jurnal Akuntansi, Manajemen Dan Ekonomi, 20*(2), 38–49. https://doi.org/10.32424/1.jame.2018.20.2.1120
Hair, J. F., Ringle, C. M., & Sarstedt, M. (2013). Partial least squares structural equation Modeling: Rigorous applications, better results and higher acceptance. *Long Range Planning, 46*(1–2), 1–12.
Heinberg, M., Katsikeas, C. S., Ozkaya, H. E., & Taube, M. (2020). How nostalgic brand positioning shapes brand equity: Differences between emerging and developed markets. *Journal of the Academy of Marketing Science, 48*(5), 869–890. https://doi.org/10.1007/s11747-019-00637-x
Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science, 43*(1), 115–135. https://doi.org/10.1007/s11747-014-0403-8
Henseler, J., Ringle, C. M., & Sarstedt, M. (2016). Testing measurement invariance of composites using partial least squares. *International Marketing Review, 33*(3), 405–431. https://doi.org/10.1108/IMR-09-2014-0304
Henseler, J., Ringle, C. M., & Sinkovics, R. R. (2009). The use of partial least squares path modeling in international marketing. *Advances in International Marketing, 20*, 277–319. https://doi.org/10.1108/S1474-7979(2009)0000020014
Javor, A., Koller, M., Lee, N., Chamberlain, L., & Ransmayr, G. (2013). Neuromarketing and consumer neuroscience: Contributions to neurology. *BMC Neurology, 13*, 1–12. https://doi.org/10.1186/1471-2377-13-13
Johansson, J. K., & Leigh, L. (2011). The rate of penetration by multinationals into emerging markets: Evidence from BRIC. *Multinational Business Review, 19*(3), 272–289. https://doi.org/10.1108/15253831111172694
Josiassen, A., Assaf, A. G., & Karpen, I. O. (2011). Consumer ethnocentrism and willingness to buy: Analyzing the role of three demographic consumer characteristics. *International Marketing Review, 28*(6), 627–646. https://doi.org/10.1108/02651331111181448
Kahneman, D. (2014). *Thinking fast and slow*. Penguin Random House Grupo Editorial.
Kaldeen, M., & Sabraz Nawaz, S. (2019). Knowledge management strategic enablers in higher education institutions in Sri Lanka. *International Journal of Grid and Distributed Computing, 12*(3), 190–198. http://ir.lib.seu.ac.lk/handle/123456789/4333.
Karoui, S., & Khemakhem, R. (2019). Consumer ethnocentrism in developing countries. *European Research on Management and Business Economics, 25*(2), 63–71. https://doi.org/10.1016/j.iedeen.2019.04.002
Kautish, P., Khare, A., & Sharma, R. (2020). The influence of values, brand consciousness, and behavioral intentions on predicting luxury fashion consumption. *The Journal of Product and Brand Management*. https://doi.org/10.1108/JPBM-08-2019-2535
Kilders, V., Caputo, V., & Liverpool-Tasie, L. S. O. (2021). Consumer ethnocentric behavior and food choices in developing countries: The case of Nigeria. *Food Policy, 99*, Article 101973. https://doi.org/10.1016/j.foodpol.2020.101973
Kurdi, B., Carroll, T. J., & Banaji, M. R. (2021). Specificity and incremental predictive validity of implicit attitudes. *Cognitive Research: Principles and Implications, 6*(1). https://doi.org/10.1186/s41235-021-00324-y
Lecerf, M., & Omrani, N. (2020). SME internationalization: The impact of information technology and innovation. *Journal of the Knowledge Economy, 11*(2), 805–824. https://doi.org/10.1007/s13132-018-0576-3
Lee, N., Chamberlain, L., & Brandes, L. (2018). Welcome to the jungle! *European Journal of Marketing, 52*(1–2), 4–38. https://doi.org/10.1108/EJM-02-2017-0122
Lewinski, P., Fransen, M. L., & Tan, E. S. (2014). Predicting advertising effectiveness by facial expressions. *Journal of Neuroscience, Psychology, and Economics, 7*(1), 1–14.
Livneh, H. (1982). On the origins of negative attitudes towards people with disabilities. *Rehabilitation Literature, 43*(11–12), 338–347.
Loijens, L., & Krips, O. (2021). *FaceReader: Methodology note*. Noldus Information Technology.
Maksan, M., Kovačić, D., & Cerjak, M. (2019). The influence of consumer ethnocentrism on purchase of domestic wine: Application of the extended theory of planned behaviour. *Appetite, 142*, 104393. https://doi.org/10.1016/j.appet.2019.104393
Maró, Z. M., Maró, G., Jámbor, Z., Czine, P., & Török, Á. (2023). Profiling the consumers of farmers' markets: A systematic review of survey-based empirical evidence. *Renewable Agriculture and Food Systems, 38*. https://doi.org/10.1017/S1742170523000467
Martínez-Ávila, M., & Fierro-Moreno, E. (2018). Aplicación de la técnica PLS-SEM. *Revista Iberoamericana para la Investigación y el Desarrollo Educativo, 8*(16). https://doi.org/10.23913/ride.v8i16.336
Ngoc, T., Nguyen, D., Dang, P. N., Tran, H., Thi, T., Nguyen, T., & Hoang Tran, P. (2022). The impact of consumer ethnocentrism on purchase intention. *Journal of Asian Finance, 9*(2), 427–436. https://doi.org/10.13106/jafeb.2022.vol9.no2.0427
Nisbett, R. E., & Wilson, T. D. (1977). Telling more than we can know. *Psychological Review, 84*(3), 231–259. https://doi.org/10.1037/0033-295X.84.3.231
Nunnally, J. C. (1978). Cronbach's Alpha. *Journal of Consumer Research, 16*, 297–334.
Palos-Sanchez, P. R., Baena-Luna, P., & Silva-O'Connor, D. (2023). Exploring employees' beliefs regarding virtual worlds. *Multimedia Tools and Applications, 82*, 24943–24965. https://doi.org/10.1007/s11042-022-14308-7
Pieters, R., Warlop, L., & Wedel, M. (2007). Breaking through the clutter. *Management Science, 53*(6), 981–995.
Pieters, R., Wedel, M., & Zhang, J. (2007). Optimal feature advertising design. *Management Science, 53*(11), 1815–1828. https://doi.org/10.1287/mnsc.1070.0732
Plassmann, H., Venkatraman, V., Huettel, S., & Yoon, C. (2015). Consumer neuroscience. *Journal of Marketing Research, 52*(4), 427–435. https://doi.org/10.1509/jmr.14.0048
Poelmans, E., Vanderlinden, M., & Rousseau, S. (2022). Ethnocentrism and wine selection. *Journal of Marketing Communications*, 1–24. https://doi.org/10.1080/13527266.2022.2127841
Rambocas, M., & Mahabir, J. M. (2021). Cultural sensitivity and ethnocentrism. *Journal of Fashion Marketing and Management, 25*(4), 644–664. https://doi.org/10.1108/JFMM-03-2020-0049
Ramsøy, T. Z. (2019). *Introduction to neuromarketing & consumer neuroscience*. Neurons Inc.
Rašković, M., Ding, Z., Hirose, M., Žabkar, V., & Fam, K. S. (2019). Segmenting young-adult consumers. *Journal of Business Research*. https://doi.org/10.1016/j.jbusres.2019.04.013
Reyes-Menendez, A., Palos-Sanchez, P., Saura, J. R., & Santos, C. R. (2022). Perceived social value. *European Management Journal, 40*(2), 224–233. https://doi.org/10.1016/j.emj.2021.06.006
Rey-Moreno, M., Felício, J. A., Medina-Molina, C., & Rufín, R. (2018). Adopting e-government. *Journal of Business Research, 88*, 542–549. https://doi.org/10.1016/j.jbusres.2017.11.015
Sabol, M., Hair, J., Cepeda, G., Roldán, J. L., & Chong, A. Y. L. (2023). PLS-SEM in information systems. *Industrial Management & Data Systems, 123*(12), 2997–3017. https://doi.org/10.1108/IMDS-07-2023-0429
Samiee, S., Shimp, T., & Sharma, S. (2005). Brand origin recognition accuracy. *Journal of International Business Studies, 36*(4), 379–397.
Sarstedt, M., Henseler, J., & Ringle, C. M. (2011). Multigroup analysis in PLS. *Advances in International Marketing, 22*, 195–219.
Sarstedt, M., Ringle, C. M., Henseler, J., & Hair, J. F. (2014). On the emancipation of PLS-SEM. *Long Range Planning, 47*(3), 154–160.
Saura, J. R., Palos-Sanchez, P., & Blanco-González, A. (2020). Collaborative CRMs. *Journal of Business & Industrial Marketing, 35*(3), 470–482. https://doi.org/10.1108/JBIM-12-2018-0412
Saura, J. R., Palos-Sánchez, P., & Velicia-Martín, F. (2020). What drives volunteers to accept a digital platform that supports NGO projects? *Frontiers in Psychology, 11*, Article 429. https://doi.org/10.3389/fpsyg.2020.00429
Septianto, F., Chiew, T. M., & Thai, N. T. (2020). Product emotional appeal. *Journal of Retailing and Consumer Services, 52*, Article 101916. https://doi.org/10.1016/j.jretconser.2019.101916
Shankarmahesh, M. (2006). Consumer ethnocentrism. *International Marketing Review, 23*(2), 146–172. https://doi.org/10.1108/02651330610660065
Sharma, P. (2015). Consumer ethnocentrism. *Journal of International Business Studies, 46*(3), 381–389. https://doi.org/10.1057/jibs.2014.42
Shimp, T. A., & Sharma, S. (1987). Consumer Ethnocentrism. *Journal of Marketing Research, 24*(3), 280–289. https://doi.org/10.2307/3151638
Shin, E., Chambers, E., & Lee, J. (2025). Korean consumers' perception of common Korean foods from different origins based on ethnocentrism. *Food Quality and Preference, 130*, 105551. https://doi.org/10.1016/j.foodqual.2025.105551
Siamagka, N. T., & Balabanis, G. (2015). Revisiting consumer ethnocentrism. *Journal of International Marketing, 23*(3), 66–86. https://doi.org/10.1509/jim.14.0085
Sorensen, H. (2016). *Inside the mind of the shopper*. Pearson Education.
Stere, S., & Trajani, B. (2015). Consumer ethnocentrism literature. *Social Sciences and Education Research Review, 2*(1), 41–54.
Sun, Y., Gonzalez-Jimenez, H., & Wang, S. (2021). E-WOM and ethnocentrism. *Journal of Business Research, 130*, 564–573. https://doi.org/10.1016/j.jbusres.2019.09.040
Thornton, H. C., & Zhao, Y. (2025). Sharing economy companies' internationalization. *International Business Review, 34*(2), Article 102355. https://doi.org/10.1016/j.ibusrev.2024.102355
Tobii Pro AB. (2019). *Tobii Pro lab user's manual*. Tobii Pro AB.
Totzek, D., & Jurgensen, G. (2020). *Sequential price disclosure*. https://doi.org/10.1002/mar.21426
Trivedi, S. D., Tapar, A. V., & Dharmani, P. (2024). Consumer ethnocentrism and product evaluation. *Journal of International Consumer Marketing, 36*(1), 41–61.
Van Loo, E. J., Grebitus, C., & Roosen, J. (2019). Origin labeled cheese. *Food Quality and Preference, 78*, Article 103716. https://doi.org/10.1016/j.foodqual.2019.05.016
Venkatraman, V., Clithero, J. A., Fitzsimons, G. J., & Huettel, S. A. (2015). Brand preferences. *Journal of Consumer Psychology, 25*(3), 458–465.
Weber, M., Lambert, J., Conrad, K., Jennings, S., & Mastal, J. (2018). Cultural system and ethnocentrism. *Systemic Practice and Action Research, 31*(6), 617–636. https://doi.org/10.1007/s11213-018-9444-0
Wedel, M., & Pieters, R. (2008a). Eye tracking for visual marketing. *Foundations and Trends in Marketing, 1*(4), 231–320.
Wedel, M., & Pieters, R. (2008b). A review of eye-tracking research in marketing. *Review of Marketing Research, 4*, 123–147. https://doi.org/10.1108/S1548-6435(2008)0000004009
Zdravković, S., & Gašević, D. (2022). Acceptance of foreign brands. *Economic Analysis, 55*(1), 48–62. https://doi.org/10.28934/ea.22.55.1.pp48-62